Rethinking Uncertainty Quantification in Diffusion-Bridge Medical Image

Translation: Pitfalls, Baselines, and Boundaries (system: TrustBridge — a training-free trust layer with Convergence Residual Fields)

Code and weights: https://github.com/ASHDUSK/TrustBridge-

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ABSTRACT

Two pitfalls quietly corrupt most uncertainty-quantification (UQ) claims made for generative medical image translation. First, *background air pixels*: they are trusted and near-zero-error, so pooling them with body pixels inflates whole-image UQ metrics severalfold—we show a headline localization score of 0.90 collapsing to 0.62 under a within-body protocol, and retract our own earlier numbers that benefited from it. Second, *dispersion is not error*: across an equal-budget comparison with TTA and input-noise baselines, our sampler-native signals fall to chance on the official T2→T1 weights (fused 0.57; the TTA baseline retains weak signal, 0.63), where residual error is dominated by fine texture—which signal, if any, survives is a property of the (model, direction) pair. We then ask what *does* survive. From the self-consistent recursive diffusion-bridge sampler (SelfRDB) we harvest two training-free signals: the *Convergence Residual Field* (CReF), the per-pixel discrepancy between successive recursive target estimates that the original sampler computes and discards, and *stochastic-bridge ensemble uncertainty*. Fused into a pixel-wise trust map, they deliver, under the within-body protocol: AUROC 0.82/0.76 (T1→T2/T2→T1, self-trained; 0.79/0.70 for a single chain at no extra cost), 0.62 on real tumor patients, and 0.83 *cross-institution* on never-seen subjects. A CReF-triggered routing scheme adds nothing on these models—single-chain CReF already dominates full-budget TTA—further confirming that which signal wins is model-dependent. A dedicated reproducibility analysis explains why officially released weights collapse to $\sim$15 dB on independently prepared data (paper value 31.56 dB)—structurally, not just photometrically (NCC 0.41)—and shows that the trust layer *barely notices*: the collapsed model still scores $62/100$—below the healthy model's 82, yet only REVIEW, never REJECT, and its pixel-level AUROC degrades to 0.61 (healthy 0.82); the originally reported blanket PASS ($94/100$, whole-image mean) was itself an air-pooling artifact. Self-consistency cannot certify correctness; external validation remains irreplaceable. We release all negative results—the protocol trap, the texture-dominated failure boundary, the absence of tumor localization, the miscalibration of image-level gates—as a boundary map for

the field, and recommend UQ claims in image translation be evaluated within the body, against cheap baselines, and per (model, direction) pair.

1. Introduction

Multi-contrast magnetic resonance imaging (MRI) captures complementary tissue contrast, but acquiring every clinically relevant weighting is expensive, slow, and sometimes contraindicated. Medical image translation—synthesizing an unacquired modality from an acquired one—addresses this scarcity and now achieves excellent fidelity. Denoising diffusion probabilistic models (DDPMs)[1,2] replaced GANs in this task for their training stability, and diffusion bridges connect the source and target distributions directly rather than through an isotropic-noise bottleneck. In particular, SelfRDB[3] introduced a self-consistent recursive diffusion bridge that reaches state-of-the-art multi-contrast translation. Trust, however, has not kept pace with fidelity. Because translation is ill-posed, generative models can *hallucinate*: synthesize plausible structure that is absent from the underlying anatomy. In clinical use, a radiologist cannot distinguish such fabrication from faithful synthesis, and this trust bottleneck—not image quality—is the primary obstacle to deployment. The SelfRDB authors state this gap explicitly: the uncertainty of diffusion bridges "is warranted to [be] evaluate[d]" in future work [3]. We close this gap with an observation: the self-consistent recursive sampler already computes, at every step and every pixel, the evidence needed to assess its own reliability—it simply discards it. SelfRDB reduces the discrepancy between successive recursive estimates to a single stopping scalar; the per-pixel residual field carries the spatial distribution of the model's *inability to self-consist*. Likewise, the bridge endpoint noise and generator latent variables make repeated runs differ; their dispersion marks where the model holds no stable belief. TrustBridge systematizes both signals into a pixel-wise trust map, examines the feasibility of image-level aggregation (§4.6) and the capability boundaries (§4.2, §4.8), and carries three properties that distinguish it from prior uncertainty machinery: (i) *training-free*—no model change, no labels, no extra network; (ii) *sampler-native*—signals are extracted from quantities the sampler already computes; and (iii) *pixel-level*—unlike image-level hallucination metrics[4], the output supports a radiologist's local reading. A further contribution is methodological. In the course of validation we found that *background air pixels silently inflate whole-image trust metrics*: air is trusted (uniform, easy) and error-free, so pooling it with body pixels flatters any "low trust predicts high error" score. Several numbers we initially reported (a tumor-localization AUROC of 0.87, decile separations of 35–$73\times$) were artifacts of this pooling and do not survive within-body evaluation. All trust metrics in this paper are therefore computed inside a body mask, and we document the inflation effect itself as a caution for future uncertainty evaluations (§§4).

Contributions.

1. We propose the Convergence Residual Field (CReF) and stochastic-bridge ensemble uncertainty as complementary, training-free trust signals for diffusion-bridge translation, fused into a pixel-wise trust map (§§3).
2. We introduce a body-masked evaluation protocol for pixel-wise trust metrics, quantify the whole-image inflation it corrects, and re-evaluate our signals against equal-budget dispersion baselines (TTA, input-noise): the fused signal localizes the top-error decile at AUROC $0.823/0.757$ (self-trained T1→T2/T2→T1), beating all equal-budget baselines, 0.62 on real tumor patients (our earlier 0.911 there was itself a whole-image artifact, §§4.8), and 0.83 cross-institution; on the *official* weights the same protocol flips the ranking and blinds our signals on T2→T1 (fused 0.57; TTA baseline 0.63) (§§4).
3. We report seven controlled boundary experiments (§§4) plus a model-level failure—the trust layer's blindness to a collapsed base model (§§4.2)—whose outcomes are mixed or negative: adaptive resampling, learnable aggregation, image-level regression, intensity harmonization, round-trip consistency, the official-weights T2→T1 signal failure, and the semantic-proxy null result. We use them to delineate the method's boundary: dispersion signals track error only where error is spatially structured, and learning-based aggregation overfits six-subject validation sets (§§4, §§5).

2. Related Work

2.1 Medical Image Translation

Unpaired cross-modality synthesis has evolved from adversarial translation —CycleGAN[5] and pix2pix[6] established the unpaired and paired regimes, followed by multi-contrast MRI specialists such as conditional GANs[7] and hybrid CNN–transformers (ResViT[8])—to diffusion-based models. SynDiff [9] introduced adversarial diffusion with a cyclic consistency prior for unsupervised medical translation, and I2I-Galip[10] replaced the discriminator with a vision–language model for efficiency. In parallel, bridges connect the two image distributions directly rather than through an isotropic-noise bottleneck: DDPMs[1] and their score-based generalization[11] provide the mathematical backbone, with score-based NCSN[12], improved schedules[2], image-to-image Schrödinger bridges[13], direct diffusion bridges with data consistency[14], dual implicit bridges[15], neural Schrödinger bridges[16], direct iterative inversion[17], and 3D bridge theory[18] spanning the design space. SelfRDB[3] combines a soft-prior bridge with self-consistent recursive sampling and reports state-of-the-art multi-contrast translation, while explicitly leaving *uncertainty quantification* of the bridge as an open problem. Our work builds directly on this sampler: we keep the model frozen and harvest two signals that SelfRDB discards.

2.2 Uncertainty Quantification in Medical Imaging

Per-pixel uncertainty is commonly obtained by MC-Dropout[19], deep ensembles[20], or test-time augmentation[21]; the epistemic/aleatoric taxonomy of [22] frames these as sampling-based epistemic estimates. Diffusion ensembles follow the same recipe: implicit segmentation ensembles[23] and per-pixel standard deviations across diffusion restorations[24] established that inter-sample dispersion tracks difficult image regions. We inherit this insight but differ in three ways: (i) our ensemble members are sampled from a *diffusion bridge* whose stochastic endpoints and latent variables are native to the sampler, not injected ad hoc; (ii) we pair ensemble dispersion with a second, complementary signal—the convergence residual—that is unique to self-consistent recursive samplers and costs nothing; and (iii) we fuse both into a pixel-wise trust map integrated with a QC workflow, whose image-level semantics we then audit, rather than reporting dispersion alone. Notably, concurrent bridge models actively *suppress* this stochasticity for reconstruction [25]; we argue it is an asset when the task is trust assessment, even though it is a defect for point synthesis.

2.3 Hallucination Assessment and Trustworthy Radiology AI

Closest to our goal, the Hallucination Index[4] quantifies fabricated content of generative reconstruction models as an *image-level* metric over repeated samples; MUMINS[26] conditions medical image synthesis on metadata and trains an uncertainty-aware prediction head to output a spatial uncertainty map; and the measurement-geometry analysis of [27] proves when a measurement operator can distinguish solutions, explaining hallucination from an operator-design perspective. These approaches either require repeated sampling at image granularity, architectural modification with retraining, or ex-ante operator design. TrustBridge instead derives pixel-level evidence *inside* the existing sampler, requires no architectural change or labels, and additionally supports rule-based gating: our framework is complementary to output-guardrail lifecycles such as RAISE[28] and provides the objective, pixel-level evidence that reporting frameworks like AI-RADS[29] call for, while remaining fully automatic.

Beyond sampling dispersion: semantic and pathology-aware signals.

Dispersion-style evidence is blind to errors the sampler makes *consistently*—a limitation we quantify in §§4. Three concurrent research lines attack this from the evaluation side and mark the natural next step for trust layers like ours. CRAFT[30] scores generated images against clinical criteria with a foundation-model Clinical Alignment Score, reducing hallucination-like content; ViCTr[31] makes synthesis pathology-aware with graded severity control via EWC-regularized LoRA adaptation; and DDWM[32] achieves zero-shot translation by learning only the target-domain distribution and fusing domain-invariant anatomy through wavelet decomposition. All three require either foundation models, retraining, or architectural change; they are complementary to sampler-native evidence and are discussed as future work in §6.

Method comparison.

Table 1. Uncertainty/hallucination assessment methods ("extra forwards" counts generator calls beyond a single base inference). The "no extra" entry of TrustBridge is specific to the current SelfRDB implementation — its recursion structure does not exist in vanilla DDPMs.

Method	Training	Extra model	Extra forwards	Granularity
MC-Dropout[19]	yes (retrain w/ dropout)	no	multiple	pixel/image
Deep ensembles[20]	yes (multiple models)	yes	multiple	pixel/image
TTA[21]	no	no	multiple	pixel/image
Diffusion ensembles[23,24]	no (reuse)	no	multiple	pixel/image
Hallucination Index[4]	depends	depends	multiple	image-level
TrustBridge (ours)	no	no	**none** (single chain)	pixel-level

3. Method

3.1 Preliminaries: the self-consistent recursive diffusion bridge

SelfRDB[3] constructs a bridge whose forward process starts at the target image x_0 and ends at a noise-added source image y_ϵ, with intermediate samples $x_t = \mu_{x_0,t} x_0 + \mu_{y,t} y + \sigma_t \epsilon$ where $\sigma_t^2 = \gamma s_t^2 / (\bar{s}_t^2 + s_t^2)$ grows monotonically toward the source end (*soft prior*). Reverse sampling starts at $x_T = y_\epsilon$; at every step t, a generator $G_\theta(x_t, t, y, \tilde{x}_0^{(r)})$ recursively refines a target estimate $\tilde{x}_0^{(r+1)}$ until the change between successive estimates falls below a threshold τ (SelfRDB: a 1% global mean), after which the self-consistent estimate $\tilde{x}_0^*$ drives a posterior draw of x_{t-1}. The publicly released configuration uses $T{=}10$ bridge steps and at most $R{=}2$ recursions, which makes per-slice inference interactive.

3.2 Convergence Residual Field (CReF)

Each reverse step produces a pixel-wise residual; SelfRDB keeps only its global mean as the stopping criterion and discards the spatial information. We retain the *last-recursion residual* of each captured reverse step:

$$\mathrm{CReF}(p) \;=\; \frac{1}{K_0 \cdot N} \sum_{i=1}^{N} \sum_{t \in \mathcal{T}_i} \left| \tilde{x}_0^{(r_t, i)}(p) - \tilde{x}_0^{(r_t - 1, i)}(p) \right|,$$

where $\mathcal{T}_i = \{t : \; t \le K_0, \; t \in \text{schedule}\}$ is the set of bridge steps actually captured for chain i ($K_0{=}3$ is the *time-index threshold* of the capture window, a parameter distinct from the recursion cap $R_{\max}{=}2$), r_t is the last recursion actually executed at step t (on early

stopping, the residual just before stopping), and $\tilde{x}_0^{(0)} \equiv 0$; the denominator is the fixed $K_0 N$ — with step skipping the captured set can be smaller than K_0, and this discrepancy is kept as-is rather than absorbed into the denominator (the formula matches the code pixel-for-pixel, verified by a hand calculation on a 4×4 input). Intuition: pixels the generator keeps revisiting without converging are those where its learned prior provides insufficient evidence — prime candidates for hallucinated content. We position it precisely as *a proxy of sampler-internal convergence instability*: non-convergence can also stem from numerical iteration or texture sensitivity and is not strictly epistemic uncertainty[22]. The signal requires no extra generator forwards (the recursion runs anyway; caching and normalization overheads are tabulated in the cost table). Figure 1 relates both signals to the sampling chain.

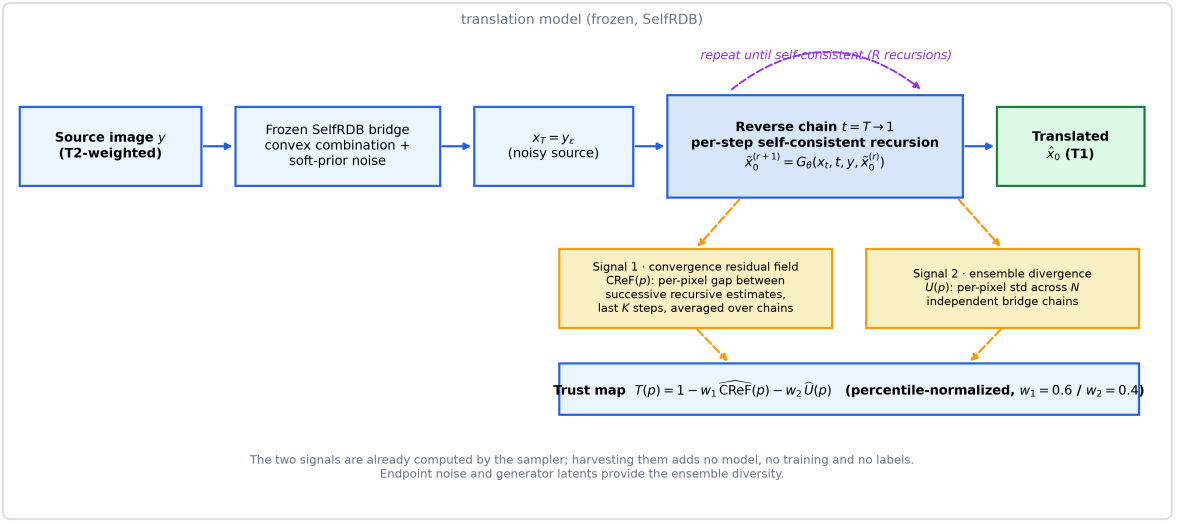


Fig. 1. Method overview. Left: the frozen SelfRDB translation model (forward bridge and self-consistent recursive reverse chain); right: the trust-quantification layer — the convergence residual field (signal 1) and multi-chain divergence (signal 2) are harvested from the recursion and fused into a pixel-wise trust map. Both signals are already computed by the sampler; harvesting them adds no model, no training and no labels.

3.3 Stochastic-bridge ensemble uncertainty

The bridge endpoint y_ϵ and the generator's internal latent variables z make two runs of the sampler on the same input differ. Running N independent chains (distinct endpoint noise and latents) and taking the per-pixel standard deviation, $U(p) = \mathrm{std}_n\{x_0^{(n)}(p)\}$, yields a second signal that measures *sampling consistency*: pixels whose synthetic value is unstable across chains are regions where the model holds no stable belief. Unlike MC-Dropout[19] or deep ensembles[20], no dropout is injected and no separate networks are trained; the dispersion arises from the bridge sampler itself. Prior diffusion-ensemble work in segmentation and restoration[23,24] established that such dispersion tracks difficult regions; we pair it with a non-dispersion signal (CReF); the pixel-level Spearman correlation between the two is measured in §§4.4, and "complementary" is used in the measured sense rather than as a conceptual assertion rather than claiming orthogonality without measurement.

3.4 Trust map fusion

Both fields are robustly normalized (99th-percentile clipping) and fused as $T(p) = 1 - w_1\widehat{\mathrm{CReF}}(p) - w_2\widehat{U}(p)$ with $w_1{=}0.6$, $w_2{=}0.4$. The feasibility of image-level aggregation (the body-region mean) is analyzed as a negative result in §§4 (image-level QC operating points); per-slice archiving (trust maps and per-step recursion statistics) is provided with the open-source repository.

4. Experiments

4.1 Setup

Datasets.

IXI[33]: 40 healthy subjects (T1/T2/PD), subject-wise split 26/6/8 with no subject overlap; $4,000+$ paired axial slices under the official protocol (per-slice statistics, official mean-norm PSNR/SSIM). **BraTS2018**[34]: 6 HGG patients (60 evaluated tumor-bearing slices) with real co-registered T2 ground truth and tumor segmentation; source FLAIR. **SynthRAD2023 pelvis**[35]: 3 patients (2 for T1w→CT, 1 for T2w→CT; 128×128 slices, 160 and 80 respectively, all prepared slices used) — a pilot-scale cross-anatomy probe rather than a population sample (§§4.10).

Models.

Two self-trained 128×128 bridges (T2→T1 and T1→T2, 30 epochs, single RTX 4060 Laptop 8 GB, recipe matched to the official release) and two self-trained pelvis bridges (T1w→CT, T2w→CT) provide the quantitative backbone; official SelfRDB weights provide 256×256 demonstration and the BraTS FLAIR→T2 evaluation.

Implementation and analysis configuration.

Table 2 freezes the single configuration used for every number in this paper; no parameter below is tuned per experiment, and deviations from it are reported as sensitivity analyses (§§4.4, §§4.10).

Table 2. Implementation and analysis configuration (one configuration for all experiments; inference seed 2026 throughout).

Component	Setting
Base model	SelfRDB bridge, 10 diffusion steps (β: $0.1 \rightarrow 3.0$)
Self-trained	128×128, 30 epochs, batch 8, lr $1.6 \cdot 10^{-4}/10^{-4}$ (G/D), Adam $(0.5, 0.9)$, bf16-mixed, seed 42, one RTX 4060 Laptop 8 GB
Sampling	N=4 independent chains, recursion cap $R_{\max}$=2, step-skip 1
Capture window	K_0=3 data-end bridge steps ($\mathcal{T}_i$=$\{t \leq K_0\}$)
Normalization	per-map 99th-percentile clip to $[0, 1]$ (fixed operator)
Fusion	$T = 1 - (0.6\,\mathrm{CReF}_n + 0.4\,U_n)$; weight set a priori
Equal budget	40 generator forwards for all N=4 dispersion signals
Image-level score	Trust Score = $100\times$ body-mean of T; PASS/REVIEW gates at 75/55 (demo defaults, uncalibrated)
Input gate	6-class modality head + per-class Mahalanobis (τ=389, fit on the gate's own training split) + flat-image rule
Statistics	per-slice metric, macro-averaged; patient-level paired bootstrap B=5000

Trust-metric protocol.

Translation quality in Table 3 follows the official whole-image mean-norm protocol for comparability with the literature. *All trust metrics* (AUROC, sparsification, decile separation, correlation) are computed *within a body mask* (source > 0.02): background air is trivially low-error and high-trust, and pooling it with body pixels inflates whole-image trust metrics severalfold. Our own official-weight runs illustrate the effect: the same fused signal scores AUROC 0.62 within the body but 0.90 on the whole image, and its whole-image sparsification gain ($+5.2$ dB) overstates the clinically meaningful within-body gain ($+0.4$ dB) by an order of magnitude. All earlier conclusions in our project that relied on whole-image trust metrics were re-evaluated; two were retracted (§§4.8, §§4.4).

Metric and mask definitions (frozen as definitions v1).

All trust metrics are computed per slice within a body mask: $M = \{\text{source} > 0.02\}$ on the per-slice min-max-normalized $[0, 1]$ source image (no morphological post-processing — a simplified protocol). Mask-threshold sensitivity is minor on brain data (within-body AUROC at thresholds $0.01/0.02/0.05$: $0.830/0.822/0.817$ for T1→T2 and $0.753/0.739/0.737$ for T2→T1) and larger on cross-anatomy data (§§4.10). The positive class is the top-10% absolute-error pixels within the body of each slice; the score is $1 - T(p)$; AUROC is computed per slice and macro-averaged (pixel-pooled and subject-macro variants in §§4.4); AUPRC uses the same positive definition with the realized positive rate reported. Δ PSNR@k% = $\mathrm{PSNR}_{\mathrm{retained}(100-k)\%} - \mathrm{PSNR}_{\mathrm{full}}$ (mean-norm, data_range = masked normalized GT maximum). The decile separation Sep is a ratio and ΔMAE its absolute-

difference counterpart — when the denominator approaches zero, Sep is an order-of-magnitude statement only (§4.5). AUSE is the per-slice macro-averaged integral of the model-minus-oracle risk–coverage difference.

Ethics.

All three datasets are public research datasets; all processing is local; no output is used for clinical decisions.

4.2 The reproducibility crisis

The collapse.

Officially released IXI weights reproduce only ~ 15 dB on independently prepared IXI data—against the paper's 31.56 dB (Table 3; in our runs the official T1→T2 weights average 12.98 dB within the body). Within the independently reconstructed preprocessing pipelines systematically tested in this study (rigid misalignment, slice z-shift, intensity windowing, cropping, orientation), the released checkpoint did not reproduce the reported performance; *the precise source of the discrepancy remains unresolved* — candidate causes including checkpoint version, training-code vs. released-weight mismatch, and dependency differences are tabulated in the reproduction checklist (docs/18, with verified/unverified status and evidence). The companion SynDiff release is independently reported to have the same problem. The failure mode is invisible: the outputs are smooth, plausible-looking images rather than noise —yet structurally wrong (within-body NCC 0.41; see below).

What the trust layer does and does not see in a collapse.

A natural hope is that sampler-native signals would "know" the base model is failing. We tested this directly. The honest answer under our corrected protocol is more graded than the one we first reported. On the collapsed official model, the body-masked trust score drops to $62.4/100$ versus 82.4 for the healthy self-trained model, the low-trust ratio rises to 41.6% (healthy: 13.7%), and pixel-level within-body AUROC degrades to 0.61 ± 0.03 (healthy: 0.82)—the signals move in the right direction, but nowhere near far enough: the structurally wrong model still lands in REVIEW (0 PASS / 18 REVIEW / 2 REJECT), never REJECT, and the originally reported blanket PASS ($94/100$, a whole-image mean) was itself inflated by trusted air. And the collapse is *not* a benign photometric offset: within-body NCC is 0.41, SSIM 29%, and a ground-truth-fitted linear intensity correction leaves PSNR at 12.3 dB—the official model synthesizes *structurally wrong but internally self-consistent* anatomy. The sampler remains largely self-consistent about its wrong output: CReF and dispersion degrade together with quality (pixel-level AUROC 0.61 versus 0.82 healthy) but never separate a wrong model from a mediocre one decisively. The lesson for deployment is starker for being graded: *training-free internal signals cannot be trusted to reject a base model that is confidently wrong*—a 20-point trust gap is a signal, not a gate. They complement, but do not

replace, external validation against ground truth; a guard layer that has never seen its base model fail cannot be trusted to recognize failure (Figure 2).

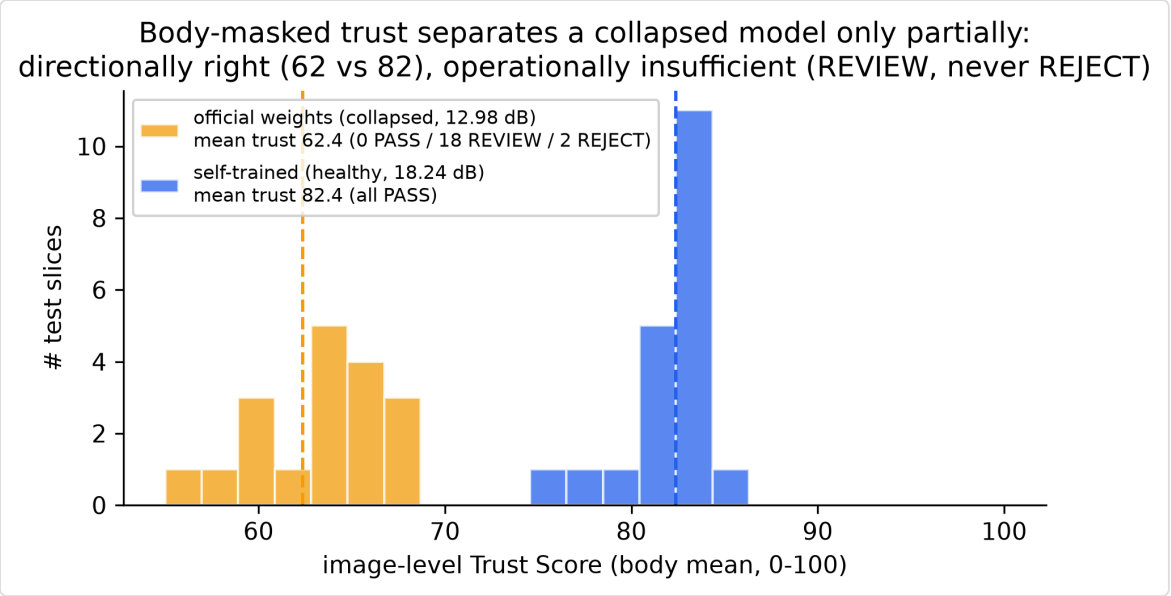


Fig. 2. Image-level trust distribution: collapsed base model vs. healthy model (20 test slices each, seed 2026). The structurally collapsed official weights (12.98 dB) score body-masked trust 62.4 (0 PASS / 18 REVIEW / 2 REJECT) versus the healthy self-trained model's 82.4 (all PASS): directionally correct, operationally insufficient. (The whole-image means we first reported—94.2 collapsed vs 90.1 healthy—were air-inflated and inverted the story.)

4.3 Translation quality

Table 3. Main results (official whole-image mean-norm PSNR/SSIM protocol). "Corrected" removes a global linear intensity mismatch whose parameters are estimated from the ground truth; it is an *error decomposition*, not a method metric—the raw column is the honest operating point. IXI: self-trained 128×128 (60 test slices each). Pelvis: self-trained, 80 test slices each. BraTS: official FLAIR→T2 weights, 60 tumor-bearing slices.

Dataset	Direction	PSNR (dB)	SSIM (%)	NCC	Corrected PSNR
IXI (paper value[3])	T2→T1	31.56	95.65	–	–
IXI, ours	T2→T1	19.81±1.10	71.0±6.6	0.884	31.74
IXI, ours	T1→T2	20.44±1.16	80.4±4.1	0.905	32.52
Pelvis, ours	T1w→CT	24.36±2.92	81.6±4.8	0.962	–
Pelvis, official	T1w→CT	21.24±2.42	64.3±6.0	0.936	–
Pelvis, ours	T2w→CT	23.13±0.83	74.5±2.7	0.961	–
Pelvis, official	T2w→CT	19.04±0.73	57.3±2.9	0.893	–
BraTS2018, official	FLAIR→T2	24.06±1.62	87.4	0.944	–

Table 3 reports translation quality. On IXI, decomposing the error shows that the content fidelity of our 128×128 self-trained models matches the paper's setting: after removing a global linear intensity mismatch, PSNR reaches 31.74/32.52 dB (T2→T1/T1→T2) versus the paper's 31.56 dB, while the raw gap is attributable to that offset and the reduced training

budget; we emphasize that the corrected column uses ground-truth-estimated parameters and serves only as a diagnosis. Our self-trained pelvis bridges beat the official weights by $+3.1/+4.1$ dB with real CT ground truth. On tumor patients, the official FLAIR→T2 weights reach 24.1 dB (whole-image protocol; 18.3 dB within the body) against real co-registered T2—a quantitative validation of bridge translation on *pathological* data in our pipeline.

4.4 Trust localization, UQ baselines, and the failure boundary

Table 4. Uncertainty-quality comparison under the within-body protocol (IXI test, 60 slices per direction, seed 2026). **Top:** self-trained 128×128 models (the quantitative backbone); all dispersion baselines use the same forward-pass budget as our ensemble (40 generator calls); CReF (N=1) uses a single chain (10 calls, *zero extra cost*). Sep is the median-across-slices ratio of mean error between least- and most-trusted deciles; for the self-trained T1→T2 model the most-trusted decile carries near-zero error, so Sep is an order-of-magnitude statement rather than a precise value. **Bottom:** the same protocol on the official weights, where the signal ranking changes and T2→T1 fails entirely. Row roles: random=control; gradient/input-noise/TTA= equal-budget baselines; CReF/ensemble/fused=ours.

	T1→T2 (self)			**T2→T1 (self)**		
Signal	AUROC	ΔPSNR@20%	Sep	AUROC	ΔPSNR@20%	Sep
Random (control)	0.497	≈0	1.0	0.500	≈0	1.0
Gradient	0.406	-0.45	0.5	0.368	-0.55	0.4
Input-noise dispersion	0.746	+1.47	65.0	0.662	+0.97	3.0
TTA dispersion	0.690	+1.33	6.2	0.640	+0.79	1.9
CReF only (N=1, no extra)	0.787	+2.28	102.0	0.705	+1.51	2.8
Ensemble only (N=4)	0.813	+2.34	159.0	0.741	+1.63	3.8
Fused (ours)	**0.823**	**+2.67**	**182.0**	**0.757**	**+1.85**	**3.7**
Same protocol on the official weights (N=4):						
Fused (official)	0.616	+0.37	1.7	0.567	+0.21	1.3
TTA (official)	0.641	+0.70	2.4	0.630	+0.95	1.5
All signals (official, T2→T1)	–	–	–	0.548–0.630 (near chance)		

Table 4 carries two messages. **On self-trained models**, the fused signal achieves AUROC $0.823/0.757$ (T1→T2/T2→T1) with $+2.67/+1.85$ dB sparsification gains. Paired subject-level bootstrap ($B=5000$, 8 subjects) shows: fused vs. ensemble alone ΔAUROC $+0.012$ (95% CI [0.003, 0.021], \emphsignificant on T1→T2}) and $+0.015$ (CI [0.003, 0.029], significant on T2→T1); fused vs. all equal-budget baselines is significant with large margins (TTA $-0.133/-0.117$, noise $-0.077/-0.095$, gradient). A single CReF chain (zero extra cost) reaches $0.787/0.705$ — most of the value at a quarter of the budget; CReF's claim is cost, not accuracy over the ensemble. **On the official weights**, the ranking changes and the T2→T1 direction collapses: our sampler-native signals fall to near-chance (fused $0.567 \pm$

0.066, ensemble 0.564, CReF 0.548) while the strongest *baseline*—TTA dispersion—retains weak signal (0.630 ± 0.088), because that model's residual error is dominated by fine texture that the convergence and bridge-dispersion measures barely track. (The "systematically below chance" values we originally reported here—e.g. fused CI $[0.442, 0.485]$—were themselves whole-image artifacts.) *The same direction, the same protocol, opposite conclusions on two models*: the failure boundary is a property of the (model, direction) pair, not of the signals, which is why we recommend validating uncertainty signals per (model, direction) before any deployment (§§5). A CReF-triggered routing scheme (uncertain regions re-examined by TTA) was tested to recover full-budget TTA quality at lower cost; on the self-trained model the curve is flat—pure CReF (0.787 at budget 10) already dominates full TTA (0.690)—so routing adds nothing there, a further confirmation that which signal wins is model-dependent.

Sensitivity to the frozen operators.

Because the fusion weight, the capture window K_0, and the 99th-percentile normalization are all fixed a priori, we verify that no conclusion depends on their precise values. Sweeping the fusion weight w_1 (the CReF share) at equal budget moves within-body AUROC by at most 0.005 across $w_1 \in [0.2, 0.6]$ ($0.826/0.827/0.822$ at $w_1{=}0.2/0.4/0.6$ on T1→T2; $0.738/0.744/0.739$ on T2→T1), with both endpoints lower (0.812 ensemble-only, 0.796 CReF-only on T1→T2). The pre-specified $w_1{=}0.6$ sits on this plateau; we report the sweep as a sensitivity analysis, not a selection. Varying the capture window $K_0 \in \{1, 2, 3, 5, 10\}$ changes AUROC by at most 0.004 ($0.820/0.822/0.822/0.821/0.818$; note $K_0{=}1$ is semantically degenerate—the residual becomes the generator output itself, as $\tilde{x}_0^{(0)} \equiv 0$—yet the metric is unchanged). On cross-anatomy pelvis data neither operator is innocuous: the same K_0 sweep spans 0.628–0.743 and the fusion-weight ranking reverses (§§4.10). The per-slice Spearman correlation between CReF and ensemble divergence is 0.806 ± 0.054 (T1→T2) versus 0.475 ± 0.077 (T2→T1): the two signals are nearly redundant where translation is easy and substantially complementary where it is hard, consistent with fusion helping mainly on T2→T1 (Table 4). Against the same top-10%-error positive class, AUPRC is $0.348 \pm 0.051/0.271 \pm 0.087$ (random ranking ≈ 0.10) and AUSE $0.014 \pm 0.004/0.031 \pm 0.013$ (T1→T2/T2→T1). Figure 3 shows the per-slice six-panel visualization: the spatial correspondence between the trust map and the true error underlies all quantitative claims.

4.5 Error ranking and decile separation

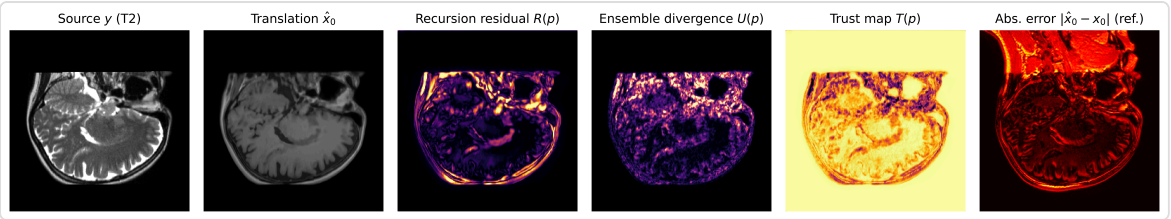


Fig. 3. Per-slice six-panel visualization (T2→T1, self-trained): source, synthesis, CReF, ensemble divergence, trust map, and the ground-truth absolute error (evaluation only). The spatial correspondence between the trust map and the error map is the core qualitative evidence: bright = trustworthy, dark = suspicious. The pattern is consistent across the 60 test slices.

Within the body, the least-trusted decile carries $182\times$ (T1→T2, median; the top three bins carry near-zero error, so the ratio is an order-of-magnitude statement), $3.7\times$ (T2→T1) and $2.1\times$ (BraTS) the mean absolute error of the most-trusted decile. These separations are far more modest than whole-image numbers suggest (35–$73\times$), because whole-image deciles are dominated by the air/body contrast. The trust map is best read as a *ranking within the body*, not an absolute error magnitude (Figure 4).

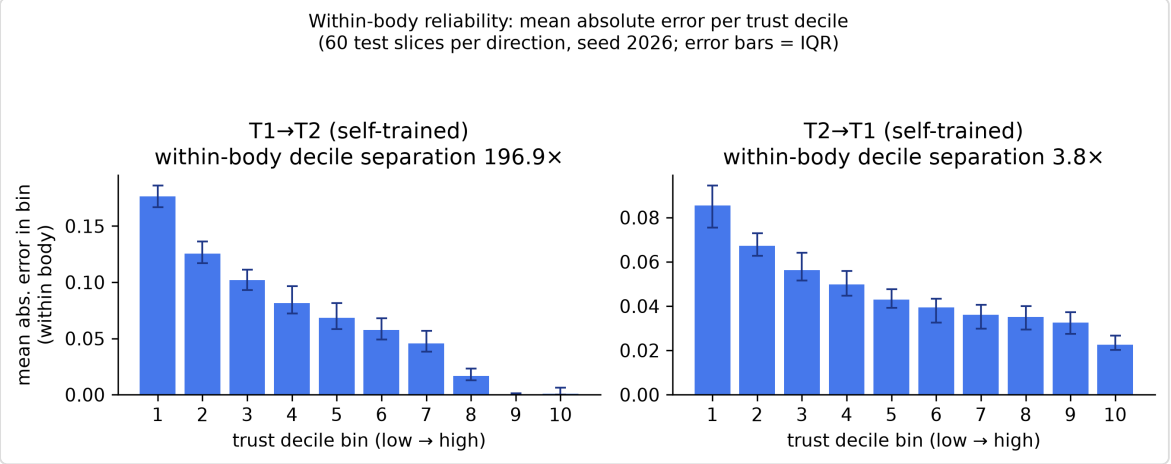


Fig. 4. Within-body reliability: mean absolute error per trust decile (60 test slices per direction, seed 2026; error bars = IQR). Both directions are near-monotone; the within-body separation is far smaller than the whole-image number (the retracted 35–$73\times$ was dominated by air/body contrast and is not used as a method metric). The top three bins carry near-zero error, where Spearman is insensitive.

4.6 Image-level QC operating points

The image level is where the layer is weakest, and we quantify this rather than hide it. Across three cohorts (IXI both directions, 60 slices each; BraTS, 60 tumor slices), the slice-level Trust Score (body mean, 0–100) correlates with slice quality (body PSNR) weakly: Pearson $r = +0.49$ (T1→T2, self-trained), $r = +0.31$ (T2→T1, self-trained), $r = -0.02$ (BraTS, official)—right-signed on the models whose signals work, absent on the one whose do not. The factory thresholds are likewise mis-calibrated to the actual score distributions: PASS$\geq$75 waves through 83% of pooled slices but catches only 18% of the bottom-quality-quartile failures, while REJECT$<$55 fires on 0.6% (with high precision, negligible recall: 2%). An empirical reading: the lowest-scoring slices are enriched for quality problems but coverage–precision trade-offs are steep (the best pooled cut, $<$55, buys 100% precision at 0.6%

coverage for the bottom-quality quartile). We therefore draw a stronger conclusion than "the thresholds need re-tuning": *global scalar trust scores should not be used as QC gates for medical image translation at all*. Slices differ so much in anatomy that no single body-mean score can separate good from bad across a cohort. We recommend the community standardize on pixel-/region-level trust evidence (the trust map, low-trust region lists) for human review, and treat any image-level gate—ours included—as a triage heuristic at best.

4.7 Boundary experiments

Seven controlled experiments delineate what the layer can and cannot do at this data scale: (1) *learnable aggregation*: a 16-feature logistic/GBDT aggregator trained on the validation split improves T2→T1 marginally ($+0.010$ AUROC) but regresses T1→T2 (-0.03), overfitting the 6-subject validation set; (2) *image-level regression*: a 16-feature ridge model predicts image PSNR worse than the baseline trust mean (0.32–0.33 vs. 0.38–0.40); (3) *adaptive resampling* and (4) *round-trip consistency signal*: no quality gain (-0.016 dB and AUROC 0.503 respectively) at 35–75% extra cost—consistent with ensemble-mean saturation; (5) *intensity harmonization*: post-hoc quantile/affine matching degrades raw PSNR (-0.8/-0.9 dB) because the per-slice intensity offset is not predictable without ground truth; and (6) \emphT2→T1 signal failure on the official weights} (§§4.4): all dispersion signals at chance when that model's residual error is texture-dominated—while the same direction works on our self-trained model (0.757), isolating the failure to the model, not the task; and (7) *semantic-proxy signals do not complement dispersion*: block-wise normalized mutual information and gradient-orientation agreement between source and synthesis reach only AUROC 0.42–0.67 where dispersion reaches 0.62–0.82, and their combination with dispersion does not improve it—structural agreement with the *source* is a weaker error surrogate than sampler-native dispersion. These results are reported in full as negative findings: at 40 subjects, the pixel-wise trust map is the output form with the most stable evidence in our experiments; its clinical utility remains unverified; reopening learnable aggregation would require a substantially larger validation set (plausibly $\geq$20 subjects, untested).

4.8 Tumor-region analysis

On the BraTS cohort (official FLAIR→T2 weights, 60 tumor-bearing slices, within-body protocol): (i) pixel error inside the tumor is $1.60\times$ the error outside (90% of slices), confirming that synthesis risk concentrates at the lesion; (ii) the trust map localizes the top-error decile with AUROC 0.624 ± 0.113 and a $+3.19$ dB sparsification gain at 20%—modest, and far below the 0.911 ± 0.037 we originally reported for this very comparison: that number was computed over the *whole image*, the same air-pooling artifact in a second guise (our second retraction); but (iii) we *retract* an earlier claim: low trust does *not* localize tumor tissue itself (within-body AUROC vs. segmentation 0.47 ± 0.11; our previously reported 0.871 was a whole-image artifact of air negatives, and mean trust is in fact *higher*

inside tumors, 0.792 vs. 0.728 outside). Flagged regions indicate where the model is unsure about its synthesis, not where pathology is. This boundary is not an implementation defect but a principled blind spot of variance-type signals. The lesson goes beyond our method: pixel-variance-style UQ measures the *variance of the synthesis process*, not the *clinical salience* of the content—a model can be highly confident (and highly consistent) about a pathologically wrong synthesis. Any UQ layer for clinical use must be audited against pathology annotations, not only against error maps (Figure 5).

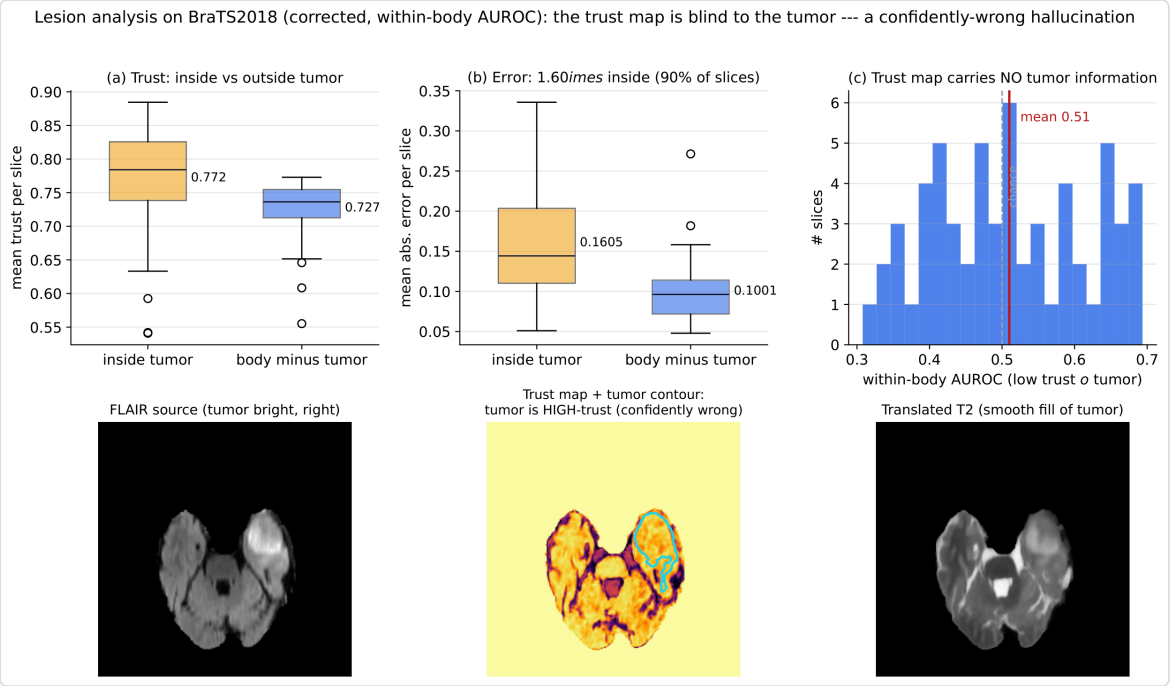


Fig. 5. Lesion blind spot (corrected, BraTS2018, 60 tumor-bearing slices, within-body protocol). (a) mean trust inside/outside tumor; (b) mean absolute error — 1.60× inside; (c) within-body AUROC of low-trust-indicates-tumor — mean 0.47, chance level (red line); (d–f) case slices: FLAIR (tumor bright), trust map + tumor contour (tumor is HIGH-trust), translation. The model confidently interpolates the homogeneous bright region into wrong content (bias-type hallucination), invisible to variance-type signals.

4.9 Stress tests: input degradation and cross-institution transfer

Input degradation.

We degrade the input along three axes—Gaussian blur ($\sigma \in \{0.5, 1, 2\}$), Gaussian noise ($\sigma \in \{0.01, 0.03, 0.06, 0.1\}$), and horizontal misalignment (2–8 px)—over the same 30 IXI test slices per dose (seed 2026), run twice: on the weak official T1→T2 weights and on the healthy self-trained weights (body PSNR 12.9 vs. 18.4 dB at zero dose; trust values are body means, a different aggregation from the whole-image score of §§4.2). The response of our trust machinery is asymmetric and instructive: the input-side Mahalanobis distance (the open-sourced input-side defense layer's OOD score, used here purely as a measurement instrument) fires dose-dependently and *identically on both models* (blur $2.9 \rightarrow 15.0$; noise $\rightarrow 9.3$; blind to small misalignment $\rightarrow 2.9$), as expected for an input-space measurement. The trust map, by contrast, is model-relative: on the healthy base, mean trust is dose-flat under blur ($0.824 \rightarrow 0.828$) and within-body localization AUROC holds ($0.82 \rightarrow 0.78$ at

the worst blur; $0.82 \rightarrow 0.83$ under noise); on the weak base, the same blur protocol *raises* mean trust ($0.63 \rightarrow 0.71$) while within-body localization sits near chance at zero dose and decays below it under misalignment ($0.61 \rightarrow 0.44$ at 8 px). A smoother input is easier for a weak sampler to be self-consistent about: knowing-what-you-don't-know at the image level is carried by the input-side detector, not the trust map, and the trust map's failure mode scales with base-model weakness (misalignment detection remains open, §§5).

Cross-institution transfer.

The strongest external objection to sampler-native trust evidence is that it may be locked to the training cohort. We therefore evaluate the identical pipeline (official FLAIR→T2 weights, identical protocol, never-seen subjects) on two held-out sets from BraTS2018 raw data: six patients from the same collection as our reference cohort (patient-level generalization) and six `CBICA` validation subjects prepared by a different institution (institution-level generalization), 120 slices each. Trust localization *transfers*: AUROC 0.707 ± 0.113 (held-out patients) and 0.834 ± 0.045 (cross-institution) versus 0.624 ± 0.113 on the reference cohort, with sparsification gains $+2.98$ and $+3.02$ dB respectively. These within-body levels are far below the $0.863/0.950/0.911$ we originally reported for the same three cohorts—all three were whole-image numbers, and correcting the protocol moved every cohort down while preserving the ordering (cross-institution strongest, held-out above reference). The caveats are real—same preprocessing recipe on our side, six subjects per set—but the direction is clear: within this cohort family, the signals are not an artifact of having seen the patients.

4.10 Cross-anatomy transfer: pelvis CT (SynthRAD)

The stronger test of sampler-native trust is not a new institution but a new anatomy and a new target modality. We run the identical frozen pipeline on the self-trained pelvis bridges (T1w→CT: 160 slices, 2 patients; T2w→CT: 80 slices, 1 patient; all prepared SynthRAD slices, including the designated validation split, with no configuration re-tuned). Slice-level results: T1w→CT reaches within-body AUROC 0.666 ± 0.061 with AUPRC 0.285 ± 0.078 (positive rate 0.10), median decile separation 1.39, and AUSE 0.037 — degraded relative to brain but still informative — whereas T2w→CT gives AUROC 0.591 ± 0.070, AUPRC 0.155 , and median decile separation 0.86: *indistinguishable from chance at the slice level*, on the hardest translation in our suite (body PSNR 15.3 dB versus 18.6 dB). Two honest observations accompany these numbers. First, the frozen operators are not anatomy-free: sweeping K_0 from 1 to 10 moves T1w→CT AUROC from 0.628 to 0.743 (the brain-tuned $K_0{=}3$ is suboptimal here), and the fusion-weight ranking *reverses* — ensemble-only attains 0.698 while CReF-only drops to 0.575, so on this (model, direction) pair the convergence residual is the weaker error carrier. Second, with 1–2 patients these are slice-level statistics without patient-level variance, and mask-threshold sensitivity is larger than on brain ($0.639/0.666/0.718$ at thresholds $0.01/0.02/0.05$). The pattern extends our failure-

boundary thesis: whether dispersion carries error information is a property of the (model, direction, anatomy) triple, and must be discovered by evaluation, not assumed — which is exactly why the per-deployment validation of §§5 comes before any use.

5. Discussion

Conditions of validity.

The trust layer works when: residual error is spatially structured, the base model has not collapsed, and evaluation is performed within the body. It fails under: texture-dominated error, bias-type hallucination (lesions), model-level collapse (§§4.2), and image-level aggregation. We carry this checklist through the discussion below.

Quality and trust calibration are separable.

Our self-trained pelvis bridges beat the official weights on PSNR by $+3.1/+4.1$ dB, yet the signal ranking changes across models (top/bottom panels of Table 4); the sharpest example is the official T2→T1 weights, whose trust signals carry no information at all. A model can be accurate and still produce uninformative uncertainty, when its residual error lacks the spatial structure that dispersion measures respond to. Uncertainty signals must be validated per (model, direction) pair before deployment—a conclusion our own negative result makes concrete.

The protocol trap.

Whole-image evaluation of pixel-wise trust metrics is systematically optimistic whenever images have trusted, low-error background—as almost all medical image translation settings do. we recommend body-masked evaluation as standard practice for uncertainty quantification in this field, and we publish our retraction trail (docs/18 and the corrected analysis scripts) as evidence that this is easy to get wrong—we did.

Positioning: a two-layer safety architecture.

Our findings split the system into two layers: an *input integrity monitor* (out-of-distribution detection, intensity anomalies, blur/noise/ misalignment — partially covered by the input-side Mahalanobis distance) and a *sampler-native trust estimator* (pixel-wise ranking by CReF and ensemble divergence); beyond both, confident structural hallucination and image-level quality degradation must be caught by external validation and human review. TrustBridge is not a complete medical-imaging safety system but an internal-signal evaluation layer within a trustworthy workflow.

Limitations.

(1) The trust map flags *unreliable* regions, not *wrong* ones: a confidently synthesized hallucination is invisible to it—and the blindness extends to the model level: a collapsed base model that is confidently wrong is flagged only for review, never rejected ($62/100$, §§4.2). (2)

Learnable aggregation overfits 6-subject validation sets; reopening it would require a substantially larger validation set (plausibly $\geq$20 subjects, untested). (3) Input-level errors (a wrong source modality) are outside the perception range of sampler-native signals; they require a dedicated input-side defense layer, which we release in the open-source repository but exclude from this paper's scope. (4) Image-level QC and tumor localization remain unsolved at this scale; the trust map additionally does not respond to input degradation—blur even raises it (§§4.9)—so degraded-input detection relies on the input-side Mahalanobis detector, and small misalignments evade both. (5) All outputs are research demonstrations and must not guide clinical decisions.

6. Conclusion

We built a pixel-wise, training-free trust-quantification layer from the discarded internals of a state-of-the-art diffusion-bridge sampler, and subjected it to a harsher evaluation than we originally planned: body-masked metrics, equal-budget dispersion baselines, and per-cohort operating-point analysis. What survives is a clear, bounded picture: on models whose errors are spatially structured (self-trained T1$\rightarrow$T2/T2$\rightarrow$T1; official weights on tumor patients), the training-free signals localize unreliable regions well (AUROC 0.62–0.83) and CReF supplies most of that value at zero cost; on texture-dominated error the signals—and every baseline we tried—retain at most weak information (best: TTA, 0.63), none usable. The broader messages are methodological: generative samplers compute more evidence than they use, and harvesting it converts a synthesis engine into a trust-assessable system; but uncertainty claims in image translation must be evaluated within the body and against cheap baselines, or they will be overstated. Future work will test whether learnable aggregation can be reopened at larger subject scale, extend the analysis to 256×256 models, and integrate radiologist feedback.

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